

From everyday LLM use to evidence about informal learning

A behavioural framework linking task ecology, turn-level codes and multi-scale association tests.

a Interaction ecology

Natural conversations in matched usage contexts.

Platform x task setting

Coding

Writing

WildChat

Copilot

User framing

Intentional or unintentional learning context

Matched settings define contextual contrasts.

b Turn-level labels

Adjacent turns coded as observable behavioural signals.

User t-1
prior state

Assistant t
support

User t+1
uptake

User codes

A active

C constructive

P passive

Assistant codes

S1 direct

S2 scaffold

S2 forms

M1 feedback

M2 hint

M3 instruct

M4 explain

M5 model

M6 question

Labels are behavioural signatures, not outcomes.

c Inference scales

Shared labels support descriptive, associational and temporal analyses.

Prevalence

Engagement signatures across settings.

Support association

Scaffolding aligned with constructive participation.

Local coupling

S2 predicts next-turn uptake conditional on prior state.
S2 -> C